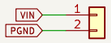
RESEARCH ONLY: fixed 99.6 V / 10 A target. No 230 V AC on this board.
Does NOT implement requested 0..100 V / 40 A laboratory PSU.

J1

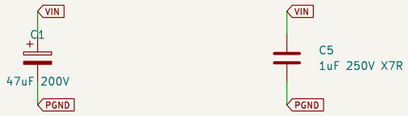


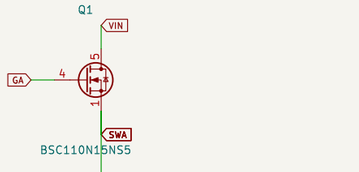
ISOLATED DC INPUT <=120V

J2

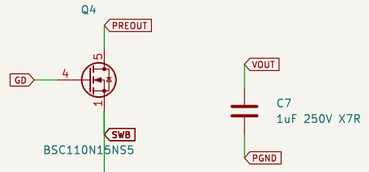


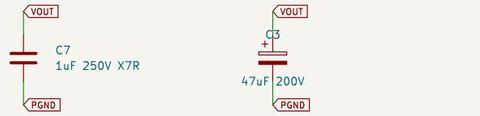
99.6V FIXED OUTPUT / 10A TARGET

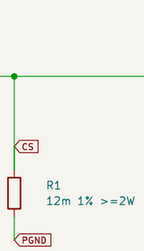


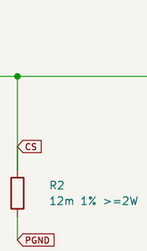


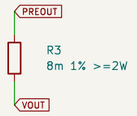


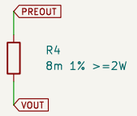












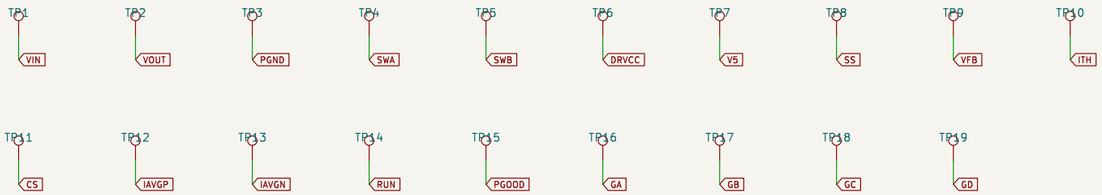
Controller and sensing



File: control.kicad_sch

6 mOhm low-side cycle shunt; 4 mOhm output average shunt.
Native CC ceiling ~12.5 A, not precision adjustable.
External 150 uH inductor: warm Isat >=25 A; vendor pending.

Guarded/differential test points. Switch-node and high-side gate signals float.
PCB CAD routing complete; see exports/power/layout_summary.json.
Hardware performance unvalidated; no Gerbers released.



Allpowercomponentsthermal/integrityreviewpending
No0Vregulation;noadjustableCC;no4-phasecurrent-sharing
External isolatedDC,<=120V;12VauxmustprecedeHV
CAD routing complete; see layout report; no hardware/fabrication qualification

Sheet: /
File: power_phase.kicad_sch

Title: RESEARCH PHASE — FIXED99.6V / 10A TARGET — NOT4kW

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 1/2

LTC3779: VIN/VOUT rated to150 V; native reference1.2 V.
12V floating auxiliary and external enable interlock required. No programmable CV/CC.

J3



AUX12 / PGND — BEFORE HV

J4



ENABLE 3.3V / SYNC / PGOOD / PGND

D1



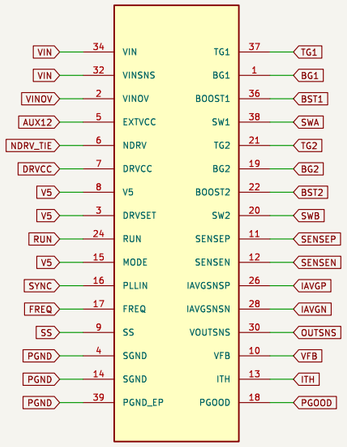
DFLS1200-7 / 1A 200V

D2



DFLS1200-7 / 1A 200V

U1



LTC3779EFE#PBF

Physical pins23/25/27/29/31/33/35 absent.
EP39 must connect PGND.
V5=5.5V. DRVCC=10V.
PGOOD pulled to5.5V.
MODE=V5: pulse skipping.
AUX failure interlock missing.

#FLG1



#FLG2



#FLG3



C9
4.7uF 16V X5R

C10
4.7uF 10V X5R

C11
1uF 50V X7R

C12
100nF 50V X7R

C13
220nF 16V X7R

C14
220nF 16V X7R

C15
220pF 100V C0G

C16
1uF 250V X7R

C17
10nF 50V X7R

C18
100pF 50V C0G

Fixed CV: 1.2*(1+820k/10k)=99.6V nominal, reference tolerance +/−1%.
VINOV:1.28*(1+998k/10k)=129.0V nominal, broad comparator tolerance; no precise OVP.

R5
0

R6
100k

R7
100

R8
100k

R9
37.4k 1%

R10
10k

R11
0

R12
0

R13
0

R14
0

R15
100

R16
100

R17
100

R18
150

R19
1k

R20
410k 0.1%

R21
410k 0.1%

R22
10k 0.1%

R23
499k 0.1%

R24
499k 0.1%

R25
10k 0.1%

R26
10k

Loopcompensationinitialonly;notverifiedatnewL/C/VOUT
ManufacturerdatasheetsforMOSFETandothercriticalpartsrequired
PinoutverifiedagainstLTC3779RevA pp2/11/12

Sheet: /Controller and sensing/
File: control.kicad_sch

Title: LTC3779 CONTROL — FIXED SETPOINT RESEARCH ONLY

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 2/2